

The orbital period T decreases as R decreases.

12

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Gravitational force provides the centripetal force.

$$\frac{GMm}{R^2} = m \frac{v^2}{R}$$

$$v = \sqrt{\frac{GM}{R}} = \sqrt{\frac{G\rho \left(\frac{4}{3}\pi R^3\right)}{R}} = \sqrt{\frac{4}{3}\pi\rho GR^2} \Rightarrow v \propto \sqrt{\rho R^2}$$

$$\frac{v_M}{v_E} = \sqrt{\frac{\rho_M R_M^2}{\rho_E R_E^2}} = \sqrt{\frac{\rho_M R_M^2}{(1.25\rho_M)(4R_M)^2}}$$

$$v_M = \left(\sqrt{\frac{1}{1.25 \times 16}}\right)(7.90)$$

$$= 1.7665 = 1.77 \text{ km s}^{-1}$$

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